Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Applications of higher-order linear differential equations - RLC Circuits

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Applications of higher-order linear differential equations - RLC Circuits
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Applications of higher-order linear differential equations - RLC Circuits

1 Applications of higher-order linear differential equations - RLC Circuits

Study focus: Applications of higher-order linear differential equations - RLC Circuits. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000238 · EV0000239

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

What is this L Cr circuit?

Key details

- IFP is measured in the current is measured in amperes and the charge and the capacitor is measured in holes.
- R is the resistance and I is the current.
- E R is equal to R into I.
- E C is equal to Q by C.
- The voltage drop across these three components is equal to the EMF supplied by the battery.
- voltage supplied by the battery is equal to voltage log across the three component that is the sum of these three voltage log is equal to electromotte force.
- VL is l into d by dt, VR is i into i, vc is equal to q by c, where i of p is the current which is flowing to the subdued and q is the charge of the capacity.
- And also we know that i is equal to dq by dt.
- Rate of change in the charge is equal to current.
- Putting this back in the differential equation, the differential equation becomes n into d square q by dt square plus r into d q by dt plus q by c is equal to e for t.
- Here's inductance resistance and capacitance icon.
- This EFT, the EMF which is supplied, is a function of P.
- I need to know, we need to know the charge.
- The current at time t is equal to zero.

- If we know the charge and current at time t is equal to zero, we get an expression for q of p and m of p.
- Find an expression for IFP.
- If R and C circuit with R the resistance, of the resistor which is present in the circuit is eight ohms, the inductance is two h capacitance is 0.15, which is connected.
- a source of voltage E of T is equal to 10 volts, assuming zero charge and zero current when T is equal to zero.
- What is the differential equation which represents the R and C circuit?
- Which is L into D square Q by D T square plus R into D Q by DT plus Q by C is equal to E of T.
- L is equal to 2 degrees.

Example or application from the lecture

- It consists of an inductor of inductance CL capacitor of capacitance C, resistor of resistance are in series with the source of EMF which is given by ESP.
- So to derive a differential equity for the LCI C2, we need to first consider the voltage drops across the three components.
- So first one is the voltage drop across the inductor, which I call it as EL, which is given by L into DIB.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000238 · EV0000239 · EV0000240 · +1 source units in pack.json

Core explanation · 05:00–09:59

R is equal to 8 ohms.

Key details

- The capacitance is equal to 01 0.15.
- And E of P is equal to it is also a constant, 10 ohms.
- B square Q by DP square plus A T to D Q by D P plus Q by 0.1 is equal to 10.
- Simplify this further.
- Nine point one has one by 10.
- Simplifying the equation further.
- Divide the entire equation by 2.
- 10 Q by 2 is 5Q is equal to 10 by 2.
- is this differential equation it's a non-homogeneous second order linear differential equation with constant coefficients we learn methods to find the solution of a non-homogeneous LB constant coefficients so what is it we know that.
- The complete solution of the non-homogeneous L V has two components.
- One is complementary function, and the other one is the particular.
- Solve the quadratic equation.
- The roots are minus 2 plus i minus 2 minus n.
- The two roots are complex in nature.
- Because there will be self-completed.

- It is e to the power alpha t into c1 cos beta t plus c2.
- Sin beta t in this case alpha is minus 2 and beta is 1.
- Minus 2t, C1 cos t plus C2 sin t.
- This doesn't alone represent the complete solution of the differential equation because the even differential equation is a non-homogeneous L D L D E.
- is a component particular integral.
- So just replace D by 0Y because 5 can be written as 5 into E to the part 0 t.
- Replace D by 0.
- which is equal to 1.
- Minus t 2 t into c1 plus t plus e2 sin t.
- This is the complementary function plus particular integral is 1.
- Q of T and E of T is equal to 0, R T is equal to 0.
- That is F0 is 0.
- Q of 0 is equal to 0.
- Use it in equation 1.
- is equal to e to the power minus 2 into 0 plus c 1 cos 0 plus c2 sine 0 plus 1 sine 0 is 0.
- C is 1.
- P found out c1.
- We'll get dq by t as e to the power minus 2 t into.
- Because it was e to the power minus 2t into C1 cos t plus c2 sin t.
- Product of two functions.
- You must product of t to the power minus 2t.
- Differentiate the second term c1 into cos t differentiation is minus sine t plus c2 into differentiation of sine is cos t.
- Plus differentiate the first term.
- P to the power minus 2p differentiation is minus 2 into e to the power minus 2t.
- Ret in the second term as it is.
- C1 cos t plus c2 sin d.
- But what is dq by dt?
- Which is equal to If P.
- DQ by DT which is equal to I of T is equal to in the place of C1, I have written the value of C.
- I of 0 is equal to.

Example or application from the lecture

- And what else is given?
- function and particular integral of this non-homogeneous linear differential equation which ref which is actually which represents the L Cr circuit which is given in the post genetic.
- Particular integral is given by phi by d square plus 4 d plus phi.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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Core explanation · 10:00–14:59

I of zero is equal to e to the power minus zero is one.

Key details

- Sin zero is zero plus c two into power zero plus minus two into e to the power zero minus one power zero plus c to sine zero.
- Simplify this you will get the value of c two as minus two.
- Substitute for c1 and c2.
- If t is equal to 5 into e to the power minus 2t sine t.
- linear differential equation to solve problems in physics and engineering, in science and engineering.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000246 · EV0000247 · EV0000248 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Applications of higher-order linear differential equations - RLC Circuits transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Applications of higher-order linear differential equations - RLC Circuits transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Applications of higher-order linear differential equations - RLC Circuits transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.
Applications of higher-order linear differential equations - RLC Circuits transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00058	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...
SRC00059	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00060	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...
SRC00061	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...

C Revision and retrieval practice